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Investigation of TMDC-based impact-ionization field effect transistor (I^2-FET) for low power logic

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ABSTRACT

To overcome the challenges of power dissipation and short-channel effects (SCEs) driven by continuous semiconductor dimensional scaling, impact-ionization field-effect transistors (I^2-FETs) based on a transition metal dichalcogenides (TMDC) material channel have emerged as highly promising candidates for next-generation low-power applications. To employ such steep-switching devices in logic applications, extensive research efforts have been continuously devoted to the I^2-FET. However, prior studies have predominantly focused on the direct-current (DC) I-V switching characteristics at the single-device level, while capacitance–voltage (C-V) characteristics — which are equally critical for circuit-level operation — have remained largely unexplored despite their growing importance. This study therefore provides a comprehensive analysis spanning both I-V and C-V characteristics of the I^2-FET and evaluates their combined impact on logic circuit performance. I-V characteristics reveal that the device exhibits an exceptionally steep subthreshold swing (SS) well below the Boltzmann limit and a high on-current, both enabled by the impact ionization mechanism, while maintaining a low off-current owing to the potential barrier at the junction. C-V characteristics exhibit anomalous gate capacitance behavior caused by an abrupt increase in carrier multiplication during device turn-on. However, circuit-level evaluation using a complementary metal-oxide-semiconductor (CMOS) inverter demonstrates for the first time that the dramatic enhancement in drive current induced by impact ionization plays a more dominant role in determining the overall switching delay than the detrimental impact of the anomalous gate capacitance. In conclusion, for the application of WSe_2 I^2-FET in integrated circuits, maximizing the drive current through the optimization of contact properties and dimensional parameters, such as the ungated length ($L_{ungated}$) and gated length (L_{gated}), are significantly more important than merely controlling the C-V characteristics. This study thus provides comprehensive design guidelines for future low-power logic circuits.

Introduction

Although continuous performance improvements in the semiconductor industry have been achieved through device scaling and increased integration density [1–3], short-channel effects and severe power dissipation have emerged as critical challenges as Si CMOS technology approaches its physical limits. As an alternative to conventional Si CMOS, transition metal dichalcogenides (TMDCs) are promising channel candidates [4–6]. The extreme atomic-level thickness of TMDCs enables superior gate controllability, thereby effectively suppressing short-channel effects. In addition, possessing carrier mobility and a bandgap comparable to those of conventional Si, TMDCs are anticipated to serve as core materials to replace conventional Si channels at sub-1 nm CMOS technology nodes [7–10].

To overcome power density limitations caused by increased integration and to reduce the supply voltage (V_{DD}), various low-power steep-slope devices exhibiting a subthreshold swing (SS) below the Boltzmann limit of 60 mV/dec at room temperature have been proposed. Representative structures actively being researched include tunneling field-effect transistors (TFETs) based on the band-to-band tunneling

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mechanism [11,12], negative-capacitance field-effect transistors (NCFETs) exploiting ferroelectric properties [13], phase-transition field-effect transistors (Phase-FETs) using phase transition phenomena [14], and impact ionization field-effect transistors (I^2-FETs) utilizing avalanche breakdown.

Among these, the I^2-FET offers the advantage of simultaneously achieving a steep SS and a high on-current through the impact ionization mechanism. However, although the conventional Si-based I^2-FET can achieve a sub-60 mV/dec SS [15], a critical drawback is the relatively high V_{DD} required to induce impact ionization [16]. Furthermore, a reliability issue in which hot carriers generated by a high electric field are injected into the gate oxide and trapped at the oxide interface, degrading the device reliability, has been identified as a critical limitation [17,18].

In contrast, TMDC materials possess a relatively low breakdown electric field compared to Si, enabling efficient impact ionization at lower voltages, and thus facilitating low-power operation [19]. Furthermore, since TMDC materials form a dangling-bond-free van der Waals (vdW) gap between the channel and the gate oxide, the generation of interface defects induced by hot-carrier injection is suppressed, thereby ensuring suppressed gate leakage and reliability [20,21]. Based on these material and structural advantages, research on TMDC channel-based I^2-FETs for low-power applications has recently attracted significant attention [22–25]. Until now, research on the TMDC-based I^2-FETs has predominantly focused on optimizing I-V characteristics and achieving a steep subthreshold swing at the single-device level. Therefore, to apply the TMDC-based I^2-FET to practical next-generation logic circuits, an understanding of the I-V characteristics as well as the C-V behavior of the device is essential. However, research on the C-V characteristics of the I^2-FET and their impact on circuit operation remains insufficient. Accordingly, this study identifies for the first time the physical mechanism of the anomalous gate capacitance peak($C_{GG,Peak}$) phenomenon occurring when the WSe_2-based I^2-FET turns on. Furthermore, by demonstrating, via CMOS inverter circuit-level evaluation, that the abrupt current increase induced by impact ionization plays a dominant role in improving circuit switching speed compared to the detrimental effect of the anomalous gate capacitance, this study presents a novel and practical paradigm for designing future low-power logic circuits.

To verify the distinctiveness of this work through comparison with prior studies, the analysis scopes of representative steep-slope low-power devices—I^2-FET, TFET, and NCFET—reported in the literature are summarized together with that of this work in Table 1. In the case of TFET and NCFET, several prior studies have comprehensively performed C-V analysis and transient inverter-level circuit evaluation in addition to DC I-V characterization [26–29]. In contrast, in the case of the I^2-FET, DC I-V characterization has been widely reported [16,24,30], whereas no prior study has reported a comprehensive analysis covering C-V characteristics and the transient inverter analysis reflecting them. This work is distinctive in that it comprehensively analyzes not only the DC I-V characteristics of the I^2-FET but also its C-V characteristics and the corresponding transient CMOS inverter operation.

Table 1
Comparison of analysis scope for steep-slope low-power devices reported in literature.

Reference	Device Type	I-V	C-V	CMOS Inverter Analysis
This Work	I^2-FET	O	O	O
[16]		O	X	O
[24]		O	X	O
[30]		O	X	X
[26]	TFET	O	O	O
[27]				
[28]	NCFET	O	O	O
[29]				

Device structure and simulation methodology

Device structure and operation principle

Fig. 1(a) shows a schematic of the I^2-FET device structure based on the WSe_2, proposed in this study. Unlike conventional MOSFET structures that solely rely on the drift–diffusion mechanism, the I^2-FET also exploits the impact ionization mechanism. To enable this mechanism, the device features a clear physical separation into a gated region and an ungated region. In the proposed structure, the gated region modulates the band bending of the adjacent ungated region through the gate fringe field and gate-drain electrostatic coupling. Furthermore, the ungated region, located between the channel and the drain, functions as an amplification region that generates a large number of electron-hole pairs driven by the strong electric field that forms when the band bending is sufficiently induced by the gate voltage.

The device operation can be summarized as follows. First, under the off-state condition where $V_{DS} > 0$ and $|V_{GS}| < |V_{TH}|$, the band bending is insufficient to trigger impact ionization in the ungated region, and thus the device stays turned off. However, as $|V_{GS}|$ increases and reaches the on-state bias ($V_{DS} > 0$, $|V_{GS}| > |V_{TH}|$), sufficient band bending is induced in the ungated region, with a highly concentrated electric field. This enhancement of the electric field and the variation in the energy band profile can be observed in the energy band diagram of Fig. 1(b). Carriers accelerated by this strong electric field gain sufficient kinetic energy to overcome the ionization threshold, thereby actively triggering impact ionization. Consequently, electron-hole pairs are multiplied significantly through the avalanche multiplication process, which rapidly increases the drain current. This effectively overcomes the fundamental 60 mV/dec SS limit of conventional MOSFETs at room temperature, while simultaneously achieving a high on-current. Additionally, during turn-on, the abrupt carrier generation in the ungated region and the charge redistribution between the gated and ungated regions can induce a transient surge in gate capacitance, which forms the physical basis of the anomalous $C_{GG,peak}$, which will be discussed in detail in the C-V analysis.

TCAD model calibration and simulation framework

To evaluate the core performance metrics—DC I-V, dynamic C-V, and inverter transient response—simulations were performed using the Sentaurus TCAD simulator from Synopsys. To model carrier mobility in the channel, the ThinLayer model was applied. This model is designed to capture the mobility degradation arising from the reduced channel thickness. Although a multilayer (thick-flake) WSe_2 channel was used, surface scattering at the oxide interface can still be significant; therefore, the ThinLayer model was adopted to capture mobility degradation under vertical gate fields. Additionally, to describe the impact ionization phenomenon, which is the core mechanism of the I^2-FET, the UniBo (University of Bologna) Avalanche model was utilized. This model reliably computes the carrier generation rate over a wide electric field range. For the I^2-FET analyzed here, a low-field gated region and a high-field ungated region coexist, and the internal electric field varies significantly between the on– and off-states. Therefore, by applying the UniBo model, which spans a wide range of electric fields from low to high, the overestimation of the generation rate in the low electric field off-state is effectively prevented, while the strong avalanche multiplication characteristics in the on-state can be accurately captured with a single continuous model. Furthermore, the source/drain electrode–WSe_2 channel junctions were modeled as Schottky contacts, and the Schottky barrier height (SBH), a key parameter investigated in this study, was set as the core sweep variable. To account for carrier generation and recombination within the device, the Shockley-Read-Hall

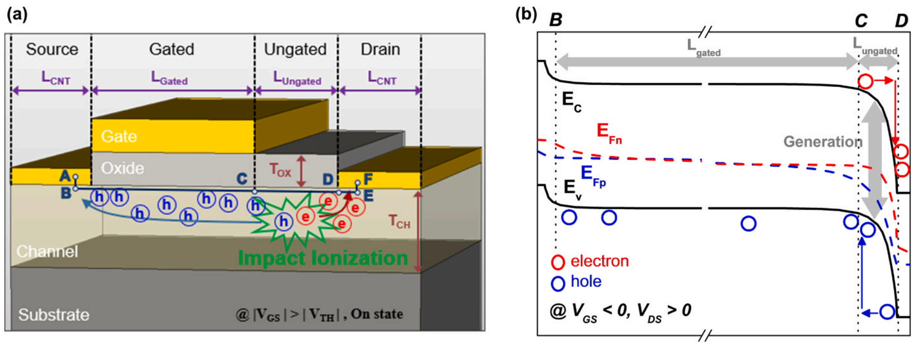

Fig. 1. Device structure and operation principle. (a) Schematic view of the Impact ionization FET (I^2-FET) structure based on the fabricated device, (b) Energy band diagram under on-state bias conditions ($V_{DS} > 0$, $|V_{GS}| > |V_{TH}|$) representing impact ionization triggered by the high electric field in the ungated region.

(SRH) and Auger recombination models were applied.

The TCAD simulation model developed in this study was constructed based on an experimentally fabricated device, the fabrication process of which is summarized as follows [20]. Fig. 2 shows a flowchart of the overall simulation and analysis methodology developed based on this fabricated device. The analysis framework of this study was divided into three main steps. The first step involves modeling and calibration. The physical structure and material property data of the experimentally fabricated device were accurately reproduced in the TCAD environment. In particular, the avalanche model coefficients were finely tuned to accurately reflect the impact ionization characteristics. Additionally, the model was calibrated to simultaneously match (i) SS, (ii) V_{TH}, and (iii) on/off ratio in the measured I–V characteristics, and the results are presented in Supplementary Fig. S1. The parameters used are summarized in Table 2. The SBH is defined by the following equation [31]:

$$\Phi_B = S(\Phi_M - \Phi_{CNL}) + (\Phi_{CNL} - \chi) \tag{1}$$

Here, S is the Fermi-level pinning factor, Φ_M is the metal work function, Φ_{CNL} is the charge neutral level of the channel, and χ is the electron affinity of the channel. The SBH range (0.485–0.92 eV) used in this study was derived from this equation. In the subsequent device-level analysis, the SBH was swept within this physical range to quantify the impact of contact property variations on I-V and C-V characteristics and circuit performance. In addition, the energy band alignment at the channel–metal interface is summarized in Supplementary Fig. S2.

The second step involves device-level evaluation. The experimentally fabricated device previously used for calibration has substantial gate dimensions, with L_{gated} of approximately 3 μm, hindering a direct assessment of its applicability in high-density circuit environments. Therefore, for practical integrated circuit-level evaluation, a reference device was defined by reducing both Lgated and $L_{ungated}$ to 50 nm and scaling V_{DD} from 3 V to 1 V. Based on this reference device, the I-V and C-V electrical characteristics were analyzed by sweeping key device

Table 2
Summary of key geometrical and material parameters used in the TCAD simulation.

Type	Parameter	Value
Dimension	Gated length (L_{gated})	70 – 10 nm
	Ungated length ($L_{ungated}$)	70 – 10 nm
	Contact length (Lcnt)	100 nm
	Channel thickness, (T_{ch})	50 nm
	Top-gate oxide thickness, (T_{ox})	20 nm
	Back-gate oxide thickness, (T_{box})	285 nm
Channel	Dielectric constant of in-plane ($\varepsilon_{\|\|}$) [50]	15.9
	Dielectric constant of out-of-plane ($\varepsilon_{\perp}$) [50]	7.8
	Energy band gap (E_g) [24]	0.94 eV
	Electron affinity (χ) [24]	4.16 eV
	Charge neutral level (CNL) [51]	4.6 eV
Oxide	Dielectric constant (ε_r)	3.9
Metal	Work function (Φ_M) [51]	5.15 eV
Metal-Semiconductor Interface	Fermi-level pinning factor (S) [51]	0.09 ~ 0.96
	Schottky barrier height (Φ_B)	0.485 ~ 0.92 eV

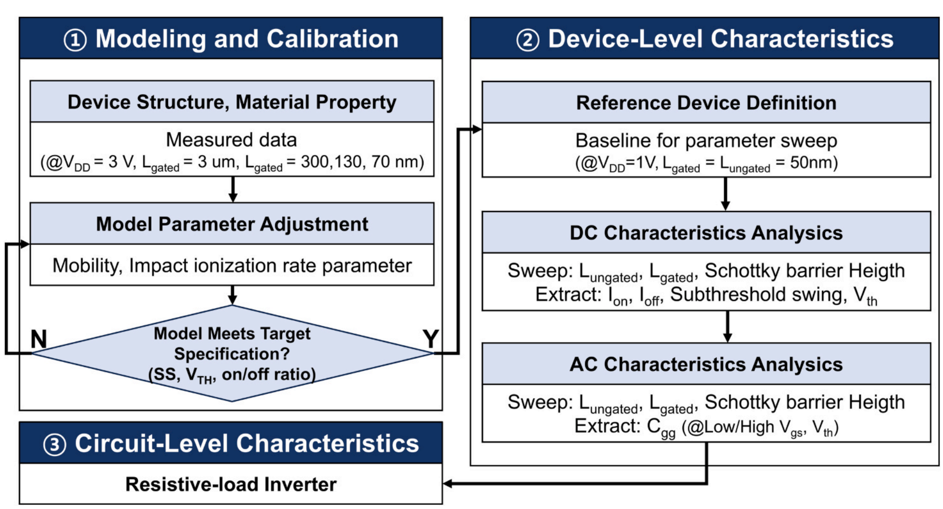

Fig. 2. Calibration process and simulation framework for WSe_2-based I^2-FETs. The overall workflow encompasses physical model calibration, device-level scaling analysis, and circuit-level performance evaluation.

parameters such as $L_{ungated}$, L_{gated}, and SBH. In particular, this step focused on analyzing the anomalous C-V characteristics observed in the I^2-FET and identifying their physical origins. In the final step, an inverter circuit-level evaluation was conducted to determine how the electrical characteristics obtained in the previous step affect practical circuit operation.

The analysis in this manuscript is based on simulations; however, for large-scale integration of the WSe_2-based I^2-FET, a wafer-scale fabrication flow is required rather than the mechanical-exfoliation-based lab-scale process. The proposed I^2-FET is expected to be realizable through a CMOS-compatible process flow as follows. The WSe_2 channel can be uniformly synthesized on a sapphire or SiO_2 substrate through MOCVD-based wafer-scale growth, with industry-level demonstration recently reported [32]. The source/drain electrodes can then be formed by photolithography and metal deposition/lift-off, fully compatible with the standard CMOS flow. Subsequently, a high-k material is deposited over the wafer by ALD to form the top gate dielectric, for which the formation of a high-k dielectric on a WSe_2 channel by ALD has been previously demonstrated [33]. Finally, the gate metal is deposited by sputtering or ALD and patterned by photolithography and etching such that it is aligned toward the source side, and this gate metal patterning step defines the asymmetric arrangement of gated and ungated regions, completing the I^2-FET structure. Meanwhile, intrinsic challenges arise in the gate dielectric formation. The absence of dangling bonds on the vdW TMDC surface hinders ALD precursor chemisorption, and the structural mismatch between the conventional amorphous high-k oxide and the crystalline TMDC channel induces interfacial charge traps that degrade device reliability [34]. To overcome these challenges, various approaches have been actively investigated, including the vdW-seeded dielectric approach using molecular crystals as seeding layers [35,36], the vdW-oxidized dielectric approach in which the TMDC itself is oxidized to form a native high-k [37] and the quasi-vdW dielectric approach in which a high-k dielectric grown on a separate substrate is transferred onto the TMDC [38,39].

Results and discussion

DC I-V characteristics of WSe_2-based I^2-FET

To analyze the I-V characteristics of the WSe_2-based I^2-FET, the DC performance was evaluated under $V_{DD} = 1$ V. The extraction criteria for key electrical metrics are defined as follows. By applying the constant current method, V_{TH} was defined as the gate voltage when I_{DS} reaches the following value, $I_{DS} = \frac{W}{L} \times 10^{-7}$A [40]. Here, L was defined as the total channel length ($L_{gated} + L_{ungated}$), and W was defined as the channel width.

Additionally, I_{ON} and I_{OFF} were extracted at $V_{GS} = V_{TH} - V_{DD}$ and $V_{GS} = V_{TH} + V_{DD}$, respectively. Unlike conventional MOSFETs, fixed voltages were not used because V_{TH} varies significantly with Lungated due to the inherent characteristics of the I^2-FET. This standardizes the evaluation conditions by maintaining a constant overdrive voltage. The subthreshold swing (SS) was extracted using the following equation, $SS = \left(\frac{\partial \log_{10} I_{DS}}{\partial V_{GS}}\right)^{-1}$ [41]. The minimum extracted SS value was designated as the minimum SS.

Fig. 3 illustrates the trends of the on-current and V_{TH} of the I^2-FET. As shown in Fig. 3(a), a reduction in $L_{ungated}$ leads to an increase in on-current and a decrease in V_{TH}. This is because decreasing $L_{ungated}$ enhances the electric field, which critically governs impact ionization rate, as expressed by $\boldsymbol{E} = \frac{V}{L_{ungated}}$. Here, V represented the voltage drop across the ungated region induced by the gate and drain biases, and **E** denoted the corresponding electric field in the ungated region. Additionally, Supplementary Fig. S3(a) shows that the impact ionization generation rate increases with the same trend as the on-current as $L_{ungated}$ decreases.

Fig. 3(b) summarizes the on-current and V_{TH} as a function of L_{gated}. A change in trend is observed around 30 nm. When L_{gated} is 30 nm or larger, the on-current increases due to a reduction in resistance. However, below 20 nm, the electric field in the gated region penetrates into the ungated region, effectively shortening $L_{ungated}$, and a V_{TH} reduction similar to that in Fig. 3(a) is observed. Fig. 3(c) summarizes the on/off ratio with respect to variations in Lungated and Lgated, referenced to the baseline device defined as $L_{ungated} = L_{gated} = 50$ nm, which serves as the reference point for all subsequent parameter sweeps in this study. The On/Off ratio map indicates that reducing $L_{ungated}$ exhibits a more pronounced impact on switching characteristics than reducing L_{gated}. This is because while the reduction of L_{gated} contributes to a decrease in channel resistance, the reduction of $L_{ungated}$ plays a dominant role by more directly enhancing the impact ionization efficiency. Additionally, Supplementary Fig. S3(b) demonstrates that the rapid increase in on-current observed with decreasing L_{gated} originates from enhanced impact ionization, as verified by the corresponding impact ionization generation rate. In summary, the turn-on of the I^2-FET is realized through the activation of impact ionization by the strong electric field formed in the ungated region and the resulting avalanche multiplication, which explosively increases the drain current. This operating mechanism can be confirmed through the impact ionization rate maps for the on– and off-states presented in Supplementary Fig. S4. In the off-state, impact ionization does not occur across the device, whereas in the on-state, strong impact ionization is observed at the drain side of the ungated region. Accordingly, the on-current is essentially determined by the electric field strength and channel resistance of the ungated region, and variations in the dimensional parameters ($L_{ungated}$, L_{gated}) that directly affect these two factors govern the on-current level.

Fig. 4 summarizes the electrical characteristics of the device as a function of SBH. Unlike conventional devices, the I^2-FET inherently suppresses drift–diffusion carrier injection from the drain due to its energy band structure, even when a gate voltage is applied. Consequently, the off-current remains flat until the device turns on. To identify the dominant factor determining this inherent leakage current level, Fig. 4(a) examines the off-current as a function of source and drain SBH.

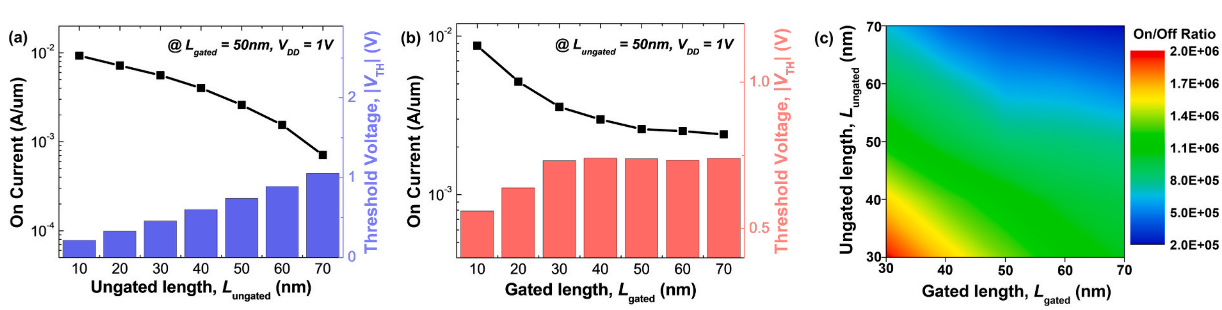

Fig. 3. I-V characteristics of the WSe_2-based I^2-FET with respect to $L_{ungated}$ and L_{gated}. (a) Variation of on-current(black line) and threshold voltage(blue bars) as a function of $L_{ungated}$. (b) Dependence of on-current(black line) and threshold voltage(red bars) on L_{gated}. (c) Contour plot mapping the on/off current ratio with respect to $L_{ungated}$ and L_{gated}.

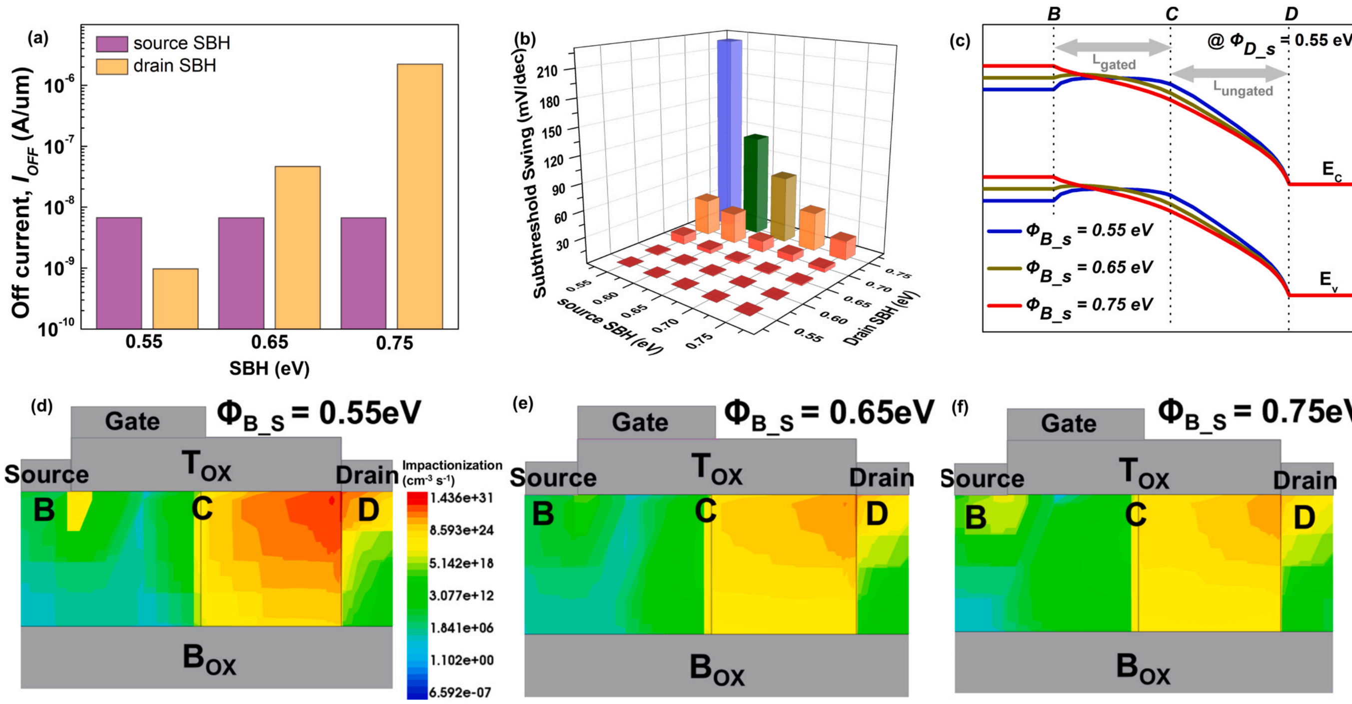

Fig. 4. I-V characteristics with respect to SBH variations. (a) variation of off-current as a function of source and drain Schottky barrier height (SBH). (b) Variation of subthreshold swing (SS) as a function of source schottky barrier hegith (SBH) (0.55–0.75 eV) and drain SBH (0.55–0.75 eV). (c) Energy band diagrams at threshold voltage with a fixed drain SBH of 0.55 eV for varying source SBH values of 0.55, 0.65, and 0.75 eV. (d–f) 2D mapping of impact ionization rates at threshold voltage corresponding to source SBH values of (d) 0.55 eV, (e) 0.65 eV, and (f) 0.75 eV.

As shown, variations in the source SBH have a negligible impact on the leakage current, whereas an increase in the drain SBH reduces the hole barrier height, leading to a substantial increase in leakage. This implies that the off-state leakage current of the I^2-FET is entirely determined by the drain SBH. Fig. 4(b) presents the SS characteristics as a function of source and drain SBH. Two possible causes of this SS degradation are identified: (i) carrier transport interference due to changes in source contact properties, and (ii) a reduction in the impact ionization rate itself. To investigate the first possibility, the energy band diagrams under the VTH condition in Fig. 4(c) were examined. Fig. 4(c) confirms that as the source SBH was reduced to 0.55 eV, a potential barrier emerged at the source-channel interface, impeding carrier transport across the junction. Subsequently, to verify the second possibility, the impact ionization rate at VTH for each source SBH condition (0.55, 0.65, and 0.75 eV) was analyzed in Fig. 4(d-f). The results reveal that the most intense and spatially extensive impact ionization occurred under the source SBH = 0.55 eV condition, where the SS was the poorest. Therefore, the SS degradation caused by source SBH variations is not attributed to a deterioration in the amplification mechanism, but rather results from the formation of a potential barrier between the source and the channel. As can be confirmed from the above analysis, the performance of the WSe_2 I^2-FET is significantly governed by the contact formation conditions, particularly by the precise control of the SBH. However, Fermi-level pinning (FLP)—in which the Fermi level of the semiconductor is pinned near the charge neutral level (CNL) due to metal-induced gap states (MIGS) formed at the metal/semiconductor junction—occurs intrinsically [31], causing the pinning factor S in Eq. (1) to decrease from 1 toward a value close to 0. When FLP is strong, the tuning of the SBH through the modulation of the metal work function becomes ineffective; in particular, when the CNL is located near the middle of the semiconductor bandgap, the SBH is pinned near the midgap, making the formation of a low-resistance contact intrinsically difficult. WSe_2 has also been reported to exhibit strong FLP, similar to other TMDCs, and considering the reported CNL position, the formation of a low SBH near the valence band, which is required for pMOS operation, is known as an intrinsic challenge [31]. To overcome this challenge, various contact engineering approaches have been actively investigated, including Fermi-level depinning and contact resistance reduction using semi-metal contacts [42], and the avoidance of direct metal/TMDC contact and mitigation of defect-induced FLP through the insertion of a 2D semiconductor interlayer [43].

The SS values reported in this study were extracted under DC measurement conditions, which is the standard approach commonly adopted in the field of low-power steep-slope device research [30,44,45]. It should be noted that, for devices with an extremely steep SS, even small noise during the switching process can affect device operation, and therefore the operational reliability of the proposed I^2-FET under noise-incorporated CMOS inverter conditions needs to be separately examined when applied to practical circuits. Given that various circuit-level techniques for mitigating the impact of noise have been actively investigated [46,47], the proposed I^2-FET is expected to achieve reliable operation when combined with such techniques in future practical circuit implementations.

Dynamic C-V characteristics of WSe_2-based I^2-FET

This subsection evaluates the C-V characteristics of the I^2-FET at V_{DS} = 1 V. The key capacitance metrics are defined as follows. $C_{GG,sat}$ is the stabilized gate capacitance under a fully formed channel, measured at $V_{GS} = V_{TH} + V_{DD}$. $C_{GG,off}$ is the off-state capacitance of the device, extracted at $V_{GS} = V_{TH} - 0.1$ V to observe the charge variation prior to the onset of impact ionization. Lastly, $C_{GG,peak}$ is the maximum C_{GG} occurring during device turn-on, observed at a $|V_{GS}|$ lower than the voltage required to reach $C_{GG,sat}$.

Fig. 5 illustrates the distinctive C-V characteristics observed in the I^2-FET and their physical mechanisms. As shown in Fig. 5(a), unlike conventional MOSFETs, the I^2-FET exhibits anomalous behavior in which C_{GG} rises sharply immediately after turn-on to form $C_{GG,peak}$, before decreasing and stabilizing at $C_{GG,sat}$. To identify the origin of this phenomenon, the space-charge distribution across each channel region was

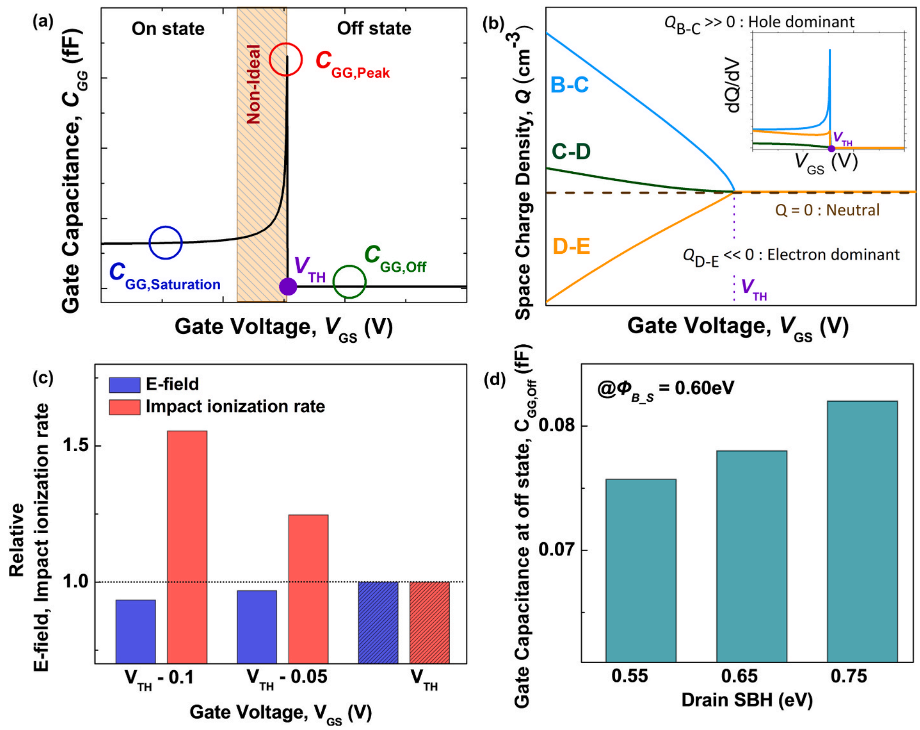

Fig. 5. Gate capacitance characteristics of the WSe_2-based I^2-FET in the off-state and turn-on region. (a) C-V characteristic of I^2-FET exhibiting an undesirable region. (b) Comparison of space charge(Q) across different regions within the channel. The inset shows dQ/dV. (c) Comparison of electric field and impact ionization rate in the post-peak V_{GS} regime, referenced to the threshold voltage V_{TH}. (d) Variation of off-state gate capacitance ($C_{GG,Off}$) as a function of Drain SBH.

analyzed in Fig. 5(b). In the ungated region, where impact ionization primarily occurs, minimal charge accumulates due to rapid carrier sweep-out driven by the intense electric field. In contrast, the effective charge forming the actual capacitance mostly resides in the gated region, confirming that the charge in this region constitutes the primary component of $C_{GG,peak}$. Fig. 5(c) explains the capacitance decrease following $C_{GG,peak}$. As the device turns on, the resistance of the ungated region diminishes, and accordingly, the voltage drop across that region reduces, leading to a weakening of the electric field applied to the ungated region. Simultaneously, the impact ionization rate continues to increase. This is because the carrier generation rate by impact ionization, G_{ii}, is governed by the following expression [48]:

$$G_{ii} = \frac{1}{q}\left(\alpha_n|J_n| + \alpha_p|J_p|\right) \tag{2}$$

$$\alpha \propto exp\left(-\frac{1}{\mathbf{E}}\right) \tag{3}$$

As indicated by the equation above, the carrier generation rate due to impact ionization is proportional to the product of the electric field and the current density. Specifically, as the device turns on, the electric field in the ungated region moderately diminishes. However, the rapidly increasing carrier density more than compensates for this decline, leading to a continuous increase in the overall impact ionization generation rate (G_{ii}). Nevertheless, the rate of increase in G_{ii} gradually slows due to the weakened electric field. This directly leads to a decrease in the effective charge variation per gate voltage (dQ/dV_{GS}), resulting in the anomalous behavior in which C_{GG} decreases after reaching $C_{GG,peak}$. Fig. 5(e) illustrates the $C_{GG,off}$ characteristics and the influence of the drain SBH. First, under the conventional off-state condition of $V_{GS} = V_{TH} - V_{DD}$, $C_{GG,off}$ remains nearly invariant regardless of drain SBH variations. This is because, in this bias region, charge modulation is negligible even when a gate voltage is applied, yielding no significant differential charge variation. In contrast, at $V_{GS} = V_{TH} - 0.1$ V, immediately prior to the onset of impact ionization, C_{GG} exhibits a clear dependence on drain SBH. As the drain SBH increases, carrier injection from the drain into the channel increases, leading to an increase in the capacitance component responding to the gate voltage just before the abrupt surge in impact ionization, resulting in an increase in C_{GG}. To clearly capture these differences in physical behavior, the $C_{GG,off}$ measurement criterion was set to $V_{GS} = V_{TH} - 0.1$ V.

Fig. 6 summarizes the C_{GG} characteristics as a function of $L_{ungated}$, L_{gated}, and drain SBH. As shown in Fig. 6(a), a reduction in $L_{ungated}$ strengthens the electric field, enhancing impact ionization and thereby increasing $C_{GG,peak}$. In contrast, $C_{GG,sat}$ decreases as the physical dimensions of the active channel are reduced. Fig. 6(b) shows that a reduction in L_{gated} leads to a linear decrease in both $C_{GG,peak}$ and $C_{GG,sat}$, owing to the reduced physical capacitor area. Notably, $C_{GG,peak}$ decreases more significantly than when $L_{ungated}$ is varied, as $C_{GG,peak}$ has an L_{gated} dependency, as previously observed in Fig. 5(b). In addition, to confirm the quantitative connection between $C_{GG,peak}$ of the proposed device and the carrier multiplication factor M induced by impact ionization in the ungated region, $L_{ungated}$ and L_{gated} sweep simulations were performed, and the detailed analysis and results are presented in Supplementary Fig. S5. Fig. 6(c) presents the C_{GG} characteristics as a function of drain SBH. As the drain SBH decreases, $C_{GG,peak}$ increases owing to enhanced energy band bending, which strengthens the internal electric field. In contrast, $C_{GG,sat}$ exhibits an opposite trend, decreasing as the drain SBH is lowered. This is attributed to carrier accumulation governed by the presence or absence of a potential barrier at the drain junction. When the drain SBH is high, carriers passing through the channel are impeded from being extracted into the drain electrode, becoming trapped by the barrier and accumulating at the channel end. These accumulated charges respond to variations in the gate voltage, forming an additional capacitance component, thereby increasing $C_{GG,sat}$. Conversely, lowering the drain SBH eliminates the barrier obstructing carrier extraction, allowing carriers to rapidly exit into the drain. Consequently, the accumulation effect is suppressed, resulting in a reduction in $C_{GG,sat}$. Fig. 6(d) presents a cross-validation of the physical mechanisms governing $C_{GG,peak}$ variation. To this end, key device

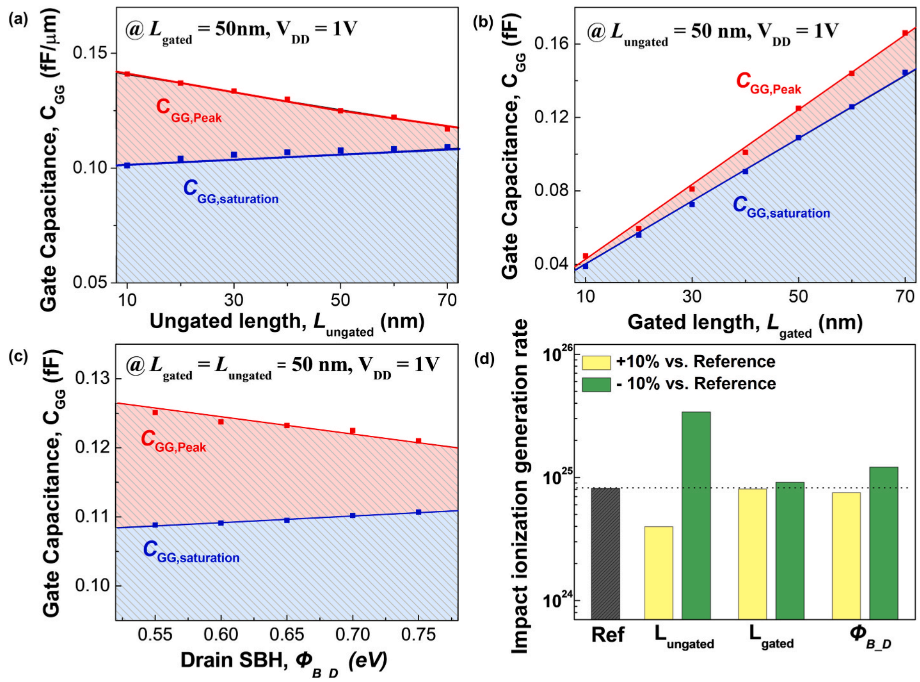

Fig. 6. C-V characteristics with respect to $L_{ungated}$, L_{gated}, and drain SBH variations. (a–c) Dependence of C_{GG} characteristics on (a) $L_{ungated}$, (b) L_{gated}, and (c) drain SBH. (d) Comparison of impact ionization generation rate under ± 10 % parameter variations, exhibiting trends consistent with the $C_{GG,Peak}$ behaviors observed in (a)–(c).

parameters—$L_{ungated}$, L_{gated}, and the drain SBH—were individually varied by 10 %, and the resulting changes in the carrier generation rate were comparatively analyzed. The analysis confirmed that the underlying causes altering $C_{GG,peak}$ are clearly distinct for each parameter. First, as $L_{ungated}$ decreased, the impact ionization rate increased significantly, closely corresponding to the previously observed surge in $C_{GG,peak}$. This demonstrates that the increase in $C_{GG,peak}$ with a shorter $L_{ungated}$ is attributed to enhanced impact ionization. Second, when L_{gated} was reduced, the impact ionization rate remained nearly unchanged, despite the observed decrease in $C_{GG,peak}$. This confirms that the decline in $C_{GG,peak}$ upon the reduction of L_{gated} is not due to a decrease in impact ionization, but rather stems from the L_{gated} dependency of $C_{GG,peak}$. Lastly, a decrease in the drain SBH leads to an increase in the impact ionization rate, owing to enhanced energy band bending in the ungated region. This verifies that the variation in $C_{GG,peak}$ caused by changes in the drain SBH is fundamentally driven by changes in the impact ionization rate.

Circuit-level performance evaluation of WSe_2-based I^2-FET

Circuit-level evaluations were performed on a CMOS inverter using mixed-mode TCAD simulations to determine the effect of the non-ideal $C_{GG,peak}$ characteristics on actual circuit operation. A conventional MOSFET, rather than an I^2-FET, was utilized for the NMOS. When the proposed I^2-FET is applied to a CMOS inverter, a key challenge arises from the mismatch between the device-level and circuit-level operating conditions. Unlike conventional pMOS, which turns on under negative drain and gate bias conditions and naturally fits the standard inverter operating range, the proposed I^2-FET turns on under positive drain bias and negative gate bias, so its turn-on condition lies outside the standard V_{IN} swing range. A similar mismatch has also been reported in impact ionization device, where the inverter was implemented using a negative V_{IN} swing range [16]. To resolve this issue, this study applied gate metal work function engineering to shift V_{TH} in the positive direction, allowing the proposed I^2-FET-based CMOS inverter to secure an appropriate switching point within the standard V_{IN} swing range without an additional negative supply voltage. To ensure a fair performance comparison between devices, all evaluations were conducted under *iso*-I_{off} conditions, with $V_{DD} = 1$ V and a load capacitance (C_{load}) of 1 fF. The inverter delay was defined as the time interval between the input voltage (V_{IN}) and output voltage (V_{OUT}) crossings at the $\frac{V_{DD}}{2}$ level. The switching frequency was calculated as frequency $= \frac{1}{\text{delay}}$.

Fig. 7 shows the variations in frequency, on-current, and $C_{GG,peak}$ as a function of $L_{ungated}$ and L_{gated}. In Fig. 7(a), when $L_{ungated}$ is reduced from 50 nm to 30 nm, $C_{GG,peak}$ increases by approximately 6.9 %, and the on-current increases by approximately 115.8 %. Meanwhile, the frequency increases by approximately 26.2 %. At first glance, these results suggest that the concurrent rise in $C_{GG,peak}$ during the $L_{ungated}$ reduction appears to limit the enhancement of switching frequency. However, an analysis of the L_{gated} variation results in Fig. 7(b) confirms that $C_{GG,peak}$ is not the dominant factor. When L_{gated} is scaled down from 50 nm to 30 nm, not only does $C_{GG,peak}$ significantly decrease by 35.2 %, but I_{ON} also increases by 38.2 % due to the reduction in channel resistance. If $C_{GG,peak}$ were the key factor determining the switching speed, these improvements would directly translate into enhanced circuit performance. However, simulation results reveal that the increase in switching frequency was limited to only 3.7 %. In conventional logic devices based on drift–diffusion carrier transport, the abrupt surge in parasitic capacitance during turn-on acts as the most critical factor causing the degradation of switching speed [49]. This confirms that $C_{GG,peak}$ is not the primary factor responsible for the degradation of switching delay. Furthermore, the disproportionately small increase in frequency relative to the I_{ON} enhancement can be attributed to the inferior performance of the conventional n-type MOSFET relative to the p-type I^2-FET. However, in the WSe_2-based I^2-FET, the explosive increase in drive current induced by impact ionization is demonstrated to overwhelm the delay penalty imposed by the anomalous $C_{GG,peak}$. This reveals that for the future design of I^2-FET-based logic circuits, maximizing the drive current through contact engineering and dimensional scaling—rather than controlling C-V characteristics—is the most critical factor in improving inverter switching performance.

Beyond the delay-centered evaluation discussed above, a joint assessment of energy and delay is also necessary to verify the suitability

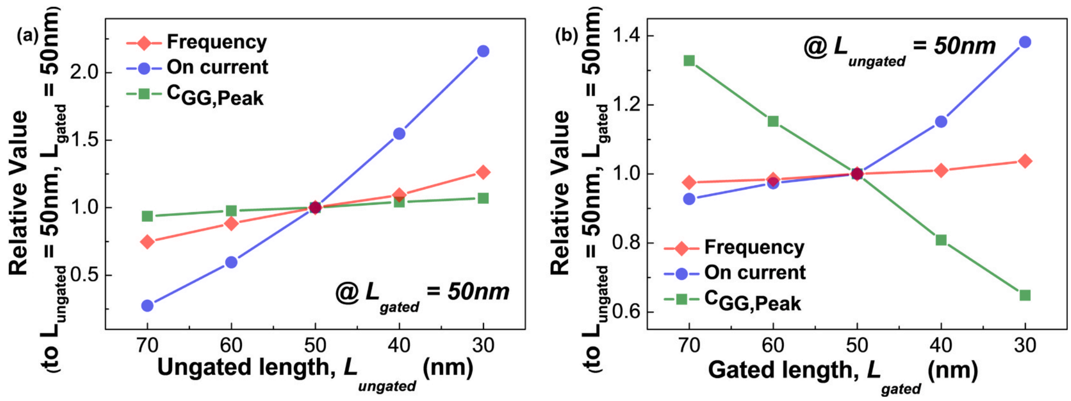

Fig. 7. Inverter circuit performance evaluation with respect to the structural dimension variations of the WSe_2-based I^2-FET. Variation of operating frequency, on-current, and $C_{GG,peak}$ as a function of (a) $L_{ungated}$ and (b) L_{gated}.

of the proposed I^2-FET for low-power logic operation. Lowering V_{DD} reduces the switching energy but also reduces the on-current and thus increases the delay, resulting in an inherent trade-off between energy and delay. Since the relative variation of Cgg under V_{DD} scaling is small in this device, the EDP behavior is governed mainly by the delay, which in turn directly reflects the on-current under V_{DD} scaling. For this reason, the on-current and V_{DD} must be examined together to identify the EDP optimum. Specifically, the V_{DD} point at which the delay begins to rise sharply as V_{DD} is reduced, or equivalently the V_{DD} point at which the delay reduction begins to slow down as V_{DD} is increased, corresponds to the EDP optimum. Moreover, although lowering V_{DD} is favorable for EDP, excessively lowering V_{DD} can limit the carrier multiplication itself and lead to a sharp drop in on-current, so an appropriate choice of V_{DD} is important. To quantitatively perform this, an EDP analysis under a V_{DD} sweep from 0.3 to 1.0 V was carried out. The analysis confirms that the EDP is minimized near $V_{DD} \approx 0.5$ V, which lies well below the typical logic operating voltage range, indicating that the proposed I^2-FET is favorable for low-power operation. The detailed results are presented in Supplementary Fig. S6.

Conclusion

A general assessment of the logic operation of a device requires both I-V and C-V characteristic analyses, the latter being directly linked to circuit operation. However, systematic C-V analysis of TMDC-based I^2-FETs has not been sufficiently performed in prior studies. In this study, the anomalous gate capacitance phenomenon occurring at the turn-on point of the WSe_2-based I^2-FET was observed for the first time, and its impact on the switching delay of a practical inverter circuit was quantitatively evaluated. In this study, the I-V and C-V characteristics of the WSe_2-based I^2-FET, along with its circuit-level performance in an inverter application, were comprehensively analyzed through TCAD simulations. The analysis of the I-V characteristics confirmed that the I^2-FET achieves a high on-current under a strong electric field via the impact ionization mechanism, while consistently maintaining a low off-current, as carrier injection is fundamentally restricted by the energy band barrier. Furthermore, this study demonstrates that optimizing the source SBH to prevent potential barrier formation at the source-channel interface allows the I^2-FET to achieve an exceptionally steep SS. The analysis of the C-V characteristics revealed that the anomalous $C_{GG,peak}$ originates from an abrupt surge in carrier multiplication when the device turns on. Interestingly, regardless of whether $C_{GG,peak}$ increases (due to $L_{ungated}$ reduction) or decreases (due to L_{gated} reduction), the magnitude of $C_{GG,peak}$ has a negligible practical impact on switching frequency. This indicates that the delay of the I^2-FET inverter is predominantly governed by the massive on-current enhancement driven by impact ionization, rather than by the variation in $C_{GG,peak}$. In other words, this study verified at the circuit level that, despite the presence of C-V non-idealities, the enhancement of on-current is the dominant factor in determining switching delay. In conclusion, integration of the WSe_2-based I^2-FET into logic circuits requires prioritizing maximizing I_{ON} through optimizing contact conditions and $L_{ungated}$ scaling, rather than controlling C-V characteristics. Recently, as the demand for low-power devices has grown significantly due to growing power consumption challenges, the I^2-FET has emerged as a highly promising candidate for next-generation electronics, offering steep switching characteristics and low leakage current. Based on the I^2-FET operating characteristics established in this study, the analysis results are expected to be applicable to the future design and optimization of various low-power logic circuits, including CMOS inverters and ring oscillators. The analysis of various parameter-dependent electrical characteristics provided in this study is expected to serve as a crucial guideline for the future design of I^2-FET-based low-power logic circuits.

CRediT authorship contribution statement

Donggi Lee: Writing – original draft, Investigation, Conceptualization. **Hanggyo Jung:** Writing – review & editing, Visualization, Methodology. **Junyeol Lee:** Software, Data curation. **Byeongjun Joo:** Visualization, Software. **Hyunbo Cho:** Validation. **Jongwook Jeon:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The following are the Supplementary data to this article: Supplementary data 1. Additional device-level analyses, including TCAD model calibration to experimental data from fabricated TMDC-based I^2-FETs, metal–TMDC contact properties and Schottky barrier height analysis, verification of impact ionization as the origin of on-current enhancement, spatial distribution of the impact ionization rate in the off- and on-states, the connection between the peak gate capacitance and the multiplication factor, and energy–delay product optimization under supply voltage scaling.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rinp.2026.108713.

Data availability

Data will be made available on request.

References

[1] Lee J, Ahn S, Lee J, Jeon J. Via-optimized CFET 6T SRAM: performance, stability, manufacturability, and scalability. IEEE Trans Electron Devices 2025;73(1):90–6. https://doi.org/10.1109/TED.2025.3637774.

[2] Kim D, Jung H, Lee J, Kim S, Jeon J. From vulnerability to robustness: Radiation-hard isolation for BPR-enabled stacked nanosheet CFETs. Nucl Eng Technol 2026;58(5):104104. https://doi.org/10.1016/j.net.2025.104104.

[3] Kim SK, Kim J, Choi G, Kwon K-W, Jeon J. A comparative analysis of middle-of-line contact architectures for complementary FETs. IEEE Access 2024;13:5396–405. https://doi.org/10.1109/ACCESS.2024.3523434.

[4] Han T, Liu H, Wang S, Chen S. Probing the electrical performance improvement of FET device based on multilayer MoS2 material. Results Phys 2023;44:106206. https://doi.org/10.1016/j.rinp.2022.106206.

[5] Abdi M, Astinchap B, Khoeini F. Electronic and thermodynamic properties of zigzag MoS2/MoSe2 and MoS2/WSe2 hybrid nanoribbons: Impacts of electric and exchange fields. Results Phys 2022;34:105253. https://doi.org/10.1016/j.rinp.2022.105253.

[6] Liu J, Sun K, Zheng X, Wang S, Lian S, Deng C, et al. Evolutions of morphology and electronic properties of few-layered MoS2 exposed to UVO. Results Phys 2020;19:103634. https://doi.org/10.1016/j.rinp.2020.103634.

[7] Kim B, Lee S, Park J-H. Innovations of metallic contacts on semiconducting 2D transition metal dichalcogenides toward advanced 3D-structured field-effect transistors. Nanoscale Horiz 2024;9(9):1417–31. https://doi.org/10.1039/D4NH00030G.

[8] Kim KS, Kwon J, Ryu H, Kim C, Kim H, Lee E-K, et al. The future of two-dimensional semiconductors beyond Moore's law. Nat Nanotechnol 2024;19(7):895–906. https://doi.org/10.1038/s41565-024-01695-1.

[9] Allain A, Kang J, Banerjee K, Kis A. Electrical contacts to two-dimensional semiconductors. Nat Mater 2015;14(12):1195–205. https://doi.org/10.1038/nmat4452.

[10] H. Jung, J. Lee, and J. Jeon. Novel vertical contact architecture for significantly reduced contact resistance in 2D nanosheet FETs, npj 2D Materials and Applications, (2025), doi: 10.1038/s41699-025-00621-w.

[11] A.M. Ionescu and H. Riel. Tunnel field-effect transistors as energy-efficient electronic switches, nature, 479, 7373, (2011), 329–337, doi: 10.1038/nature10679.

[12] Appenzeller J, Lin Y-M, Knoch J, Avouris P. Band-to-band tunneling in carbon nanotube field-effect transistors. Phys Rev Lett 2004;93(19):196805. https://doi.org/10.1103/PhysRevLett.93.196805.

[13] Salahuddin S, Datta S. Use of negative capacitance to provide voltage amplification for low power nanoscale devices. Nano Lett 2008;8(2):405–10. https://doi.org/10.1021/nl071804g.

[14] Jung H, Chang J, Jeon J. Evaluation of various phase-transition materials for steep switching super-low power application in the latest technology node. Results Phys 2024;60:107619. https://doi.org/10.1016/j.rinp.2024.107619.

[15] Gopalakrishnan K, Griffin PB, Plummer JD. Impact ionization MOS (I-MOS)-Part I: device and circuit simulations. IEEE Trans Electron Devices 2004;52(1):69–76. https://doi.org/10.1109/TED.2004.841344.

[16] Yuan B, Chen Z, Chen Y, Tang C, Chen W, Cheng Z, et al. High drain field impact ionization transistors as ideal switches. Nat Commun 2024;15(1):9038. https://doi.org/10.1038/s41467-024-53337-8.

[17] Cao W, Bu H, Vinet M, Cao M, Takagi S, Hwang S, et al. The future transistors. Nature 2023;620(7974):501–15. https://doi.org/10.1038/s41586-023-06145-x.

[18] Gopalakrishnan K, Woo R, Jungemann C, Griffin PB, Plummer JD. Impact ionization MOS (I-MOS)-part II: experimental results. IEEE Trans Electron Devices 2004;52(1):77–84. https://doi.org/10.1109/TED.2004.841345.

[19] Gao A, Lai J, Wang Y, Zhu Z, Zeng J, Yu G, et al. Observation of ballistic avalanche phenomena in nanoscale vertical InSe/BP heterostructures. Nat Nanotechnol 2019;14(3):217–22. https://doi.org/10.1038/s41565-018-0348-z.

[20] Wang Y, Sarkar S, Yan H, Chhowalla M. Critical challenges in the development of electronics based on two-dimensional transition metal dichalcogenides. Nat Electron 2024;7(8):638–45. https://doi.org/10.1038/s41928-024-01210-3.

[21] Yan H, Liu L, Loh L, Chhowalla M, Wang Y. Gate dielectrics for transistors based on two-dimensional transition metal dichalcogenide semiconductors. APL Electron. Dev. 2025;1:3. https://doi.org/10.1063/5.0275690.

[22] Elahi E, Khan MF, Aziz J, Ahsan U, Chauhan P, Assiri MA, et al. Emerging avalanche field-effect transistors based on two-dimensional semiconductor materials and their sensory applications. J Mater Chem C 2025;13(31):15767–95. https://doi.org/10.1039/D5TC02021B.

[23] Kang T, Park J, Jung H, Choi H, Lee SM, Lee N, et al. High-κ Dielectric (HfO2)/2D Semiconductor (HfSe2) Gate Stack for Low-Power Steep-Switching Computing Devices. Adv Mater 2024;36(26):2312747. https://doi.org/10.1002/adma.202312747.

[24] Choi H, Li J, Kang T, Kang C, Son H, Jeon J, et al. A steep switching WSe2 impact ionization field-effect transistor. Nat Commun 2022;13(1):6076. https://doi.org/10.1038/s41467-022-33770-3.

[25] Liu Y, Guo J, Song W, Wang P, Gambin V, Huang Y, et al. Ultra-steep slope impact ionization transistors based on graphene/InAs heterostructures. Small Struct 2021;2(1):2000039. https://doi.org/10.1002/sstr.202000039.

[26] Mookerjea S, Krishnan R, Datta S, Narayanan V. Effective capacitance and drive current for tunnel FET (TFET) CV/I estimation. IEEE Trans Electron Devices 2009;56(9):2092–8. https://doi.org/10.1109/TED.2009.2026516.

[27] Y. Morita, K. Fukuda, Y. Liu, T. Mori, W. Mizubayashi, S.-i. O'uchi, H. Fuketa, S. Otsuka, S. Migita, and M. Masahara. Tunnel FinFET CMOS inverter with very low short-circuit current for ultralow-power Internet of Things application, Japanese Journal of Applied Physics, 56, 4S, (2017), 04CD19, doi: 10.7567/JJAP.56.04CD19.

[28] D. Kwon, Y.-H. Liao, Y.-K. Lin, J.P. Duarte, K. Chatterjee, A.J. Tan, A.K. Yadav, C. Hu, Z. Krivokapic, S. Salahuddin, Response speed of negative capacitance FinFETs, in 2018 IEEE Symposium on VLSI Technology: IEEE, (2018), 49–50, doi: 10.1109/VLSIT.2018.8510626.

[29] Vega RA, Ando T, Philip TM. Junction design and complementary capacitance matching for NCFET CMOS logic. IEEE J Electron Devices Soc 2021;9:691–703. https://doi.org/10.1109/JEDS.2021.3095923.

[30] Kang C, Choi H, Son H, Kang T, Lee S-M, Lee S. A steep-switching impact ionization-based threshold switching field-effect transistor. Nanoscale 2023;15(12):5771–7. https://doi.org/10.1039/D2NR06547A.

[31] Liu X, Choi MS, Hwang E, Yoo WJ, Sun J. Fermi level pinning dependent 2D semiconductor devices: challenges and prospects. Adv Mater 2022;34(15):2108425. https://doi.org/10.1002/adma.202108425.

[32] C. Dorow, T. Schram, Q. Smets, K. O'Brien, K. Maxey, C.-C. Lin, L. Panarella, B. Kaczer, N. Arefin, A. Roy. Exploring manufacturability of novel 2D channel materials: 300 mm wafer-scale 2D NMOS & PMOS using MoS 2, WS 2, & WSe 2, in 2023 International Electron Devices Meeting (IEDM): IEEE, (2023), 1–4, doi: 10.1109/IEDM45741.2023.10413874.

[33] Pudasaini PR, Stanford MG, Oyedele A, Wong AT, Hoffman AN, Briggs DP, et al. High performance top-gated multilayer WSe2 field effect transistors. Nanotechnology 2017;28(47):475202. https://doi.org/10.1088/1361-6528/aa8081.

[34] Kim YH, Lee D, Huh W, Lee J, Lee D, Wang G, et al. Gate stack engineering of two-dimensional transistors. Nat Electron 2025;8(9):770–83. https://doi.org/10.1038/s41928-025-01448-5.

[35] Li W, Zhou J, Cai S, Yu Z, Zhang J, Fang N, et al. Uniform and ultrathin high-κ gate dielectrics for two-dimensional electronic devices. Nat Electron 2019;2(12):563–71. https://doi.org/10.1038/s41928-019-0334-y.

[36] Xu Y, Liu T, Liu K, Zhao Y, Liu L, Li P, et al. Scalable integration of hybrid high-κ dielectric materials on two-dimensional semiconductors. Nat Mater 2023;22(9):1078–84. https://doi.org/10.1038/s41563-023-01626-w.

[37] Y. Zhang, J. Yu, R. Zhu, M. Wang, C. Tan, T. Tu, X. Zhou, C. Zhang, M. Yu, and X. Gao. A single-crystalline native dielectric for two-dimensional semiconductors with an equivalent oxide thickness below 0.5 nm, Nature Electronics, 5, 10, (2022), 643–649, doi: 10.1038/s41928-022-00824-9.

[38] Huang J-K, Wan Y, Shi J, Zhang J, Wang Z, Wang W, et al. High-κ perovskite membranes as insulators for two-dimensional transistors. Nature 2022;605(7909):262–7. https://doi.org/10.1038/s41586-022-04588-2.

[39] Lu Z, Chen Y, Dang W, Kong L, Tao Q, Ma L, et al. Wafer-scale high-κ dielectrics for two-dimensional circuits via van der Waals integration. Nat Commun 2023;14(1):2340. https://doi.org/10.1038/s41467-023-37887-x.

[40] Ortiz-Conde A, Sánchez FG, Liou JJ, Cerdeira A, Estrada M, Yue Y. A review of recent MOSFET threshold voltage extraction methods. Microelectron Reliab 2002;42(4–5):583–96. https://doi.org/10.1016/S0026-2714(02)00027-6.

[41] Boucart K, Ionescu AM. Double-gate tunnel FET with high-$\kappa $ gate dielectric. IEEE Trans Electron Devices 2007;54(7):1725–33. https://doi.org/10.1109/TED.2007.899389.

[42] Wang C, Guo J, Liu D, Lin Z, Guo S, Cai S, et al. Band-hybridized selenium contact for p-type semiconductors. Nat Nanotechnol 2026;21(2):207–15. https://doi.org/10.1038/s41565-025-02084-y.

[43] Pande G, Siao J-Y, Chen W-L, Lee C-J, Sankar R, Chang Y-M, et al. Ultralow Schottky barriers in hexagonal boron nitride-encapsulated monolayer WSe2 tunnel field-effect transistors. ACS Appl Mater Interfaces 2020;12(16):18667–73. https://doi.org/10.1021/acsami.0c01025.

[44] Saeidi A, Rosca T, Memisevic E, Stolichnov I, Cavalieri M, Wernersson L-E, et al. Nanowire tunnel FET with simultaneously reduced subthermionic subthreshold swing and off-current due to negative capacitance and voltage pinning effects. Nano Lett 2020;20(5):3255–62. https://doi.org/10.1021/acs.nanolett.9b05356.

[45] Si M, Yang L, Zhou H, Ye PD. β-Ga2O3 nanomembrane negative capacitance field-effect transistors with steep subthreshold slope for wide band gap logic applications. ACS Omega 2017;2(10):7136–40. https://doi.org/10.1021/acsomega.7b01289.

[46] S. Ahmad, M.K. Gupta, N. Alam, M. Hasan, Single-ended Schmitt-trigger-based robust low-power SRAM cell, IEEE Transactions on Very Large Scale Integration (VLSI) Systems, 24, 8, (2016), 2634–2642, doi: 10.1109/TVLSI.2016.2520490.

[47] Kannan R, Rangarajan R. Low power noise immune node voltage comparison keeper design for high speed architectures. Microprocess Microsyst 2020;77:103192. https://doi.org/10.1016/j.micpro.2020.103192.

[48] Van Overstraeten R, De Man H. Measurement of the ionization rates in diffused silicon pn junctions. Solid State Electron 1970;13(5):583–608. https://doi.org/10.1016/0038-1101(70)90139-5.

[49] Lee J, Jung H, Jin D, Jeon J. Exploring optimal TMDC multi-channel GAA-FET architectures at sub-1nm nodes. J Sci: Adv Mater Devices 2025;10(3):100931. https://doi.org/10.1016/j.jsamd.2025.100931.

[50] Laturia A, Van de Put ML, Vandenberghe WG. Dielectric properties of hexagonal boron nitride and transition metal dichalcogenides: from monolayer to bulk. npj 2D Mater Appl 2018;2(1):6. https://doi.org/10.1038/s41699-020-00163-3.

[51] Liu Y, Guo J, Zhu E, Liao L, Lee S-J, Ding M, et al. Approaching the Schottky–Mott limit in van der Waals metal–semiconductor junctions. Nature 2018;557(7707):696–700. https://doi.org/10.1038/s41586-018-0129-8.